Functions- Part 2: Linear, Quadratic, Shifting Graphs

Course: Pre-Calculus-I

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Linear Functions

Definition

A **linear function** has the form:

$$f(x) = mx + b$$

where m is the slope and b is the y-intercept.

Key Properties:

- Graph is a straight line.
- Domain: $\mathbb{R}$, Range: $\mathbb{R}$ (if $m \neq 0$).
- Constant rate of change (slope m).
- $m > 0$: increasing; $m < 0$: decreasing; $m = 0$: constant.

Example: $f(x) = 2x + 1$, slope $= 2$, y-intercept $= 1$.

Quadratic Functions

Definition

A **quadratic function** has the form:

$$f(x) = ax^2 + bx + c, \quad a \neq 0$$

Key Properties:

- Graph is a parabola.
- Domain: $\mathbb{R}$.
- Range: $[k, \infty)$ if $a > 0$, $(-\infty, k]$ if $a < 0$, where k is the vertex y-value.
- Vertex: $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$.
- Axis of symmetry: $x = -\frac{b}{2a}$.
- $a > 0$: opens upward; $a < 0$: opens downward.

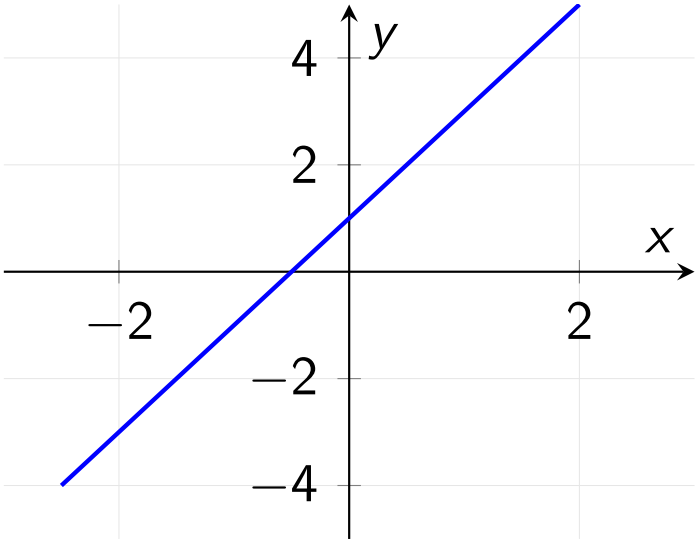
Example: $f(x) = x^2 - 4x + 3$, vertex at $(2, -1)$.

Linear vs. Quadratic Functions: Differences

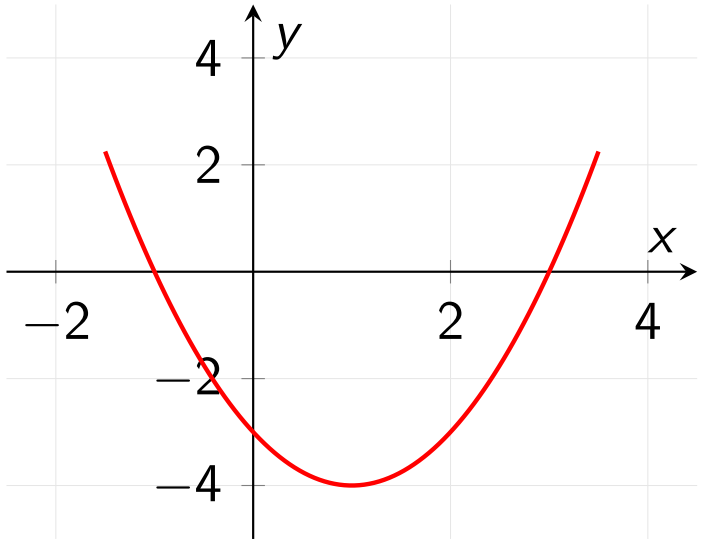
Feature	**Linear**	**Quadratic**
Form	$f(x) = mx + b$	$f(x) = ax^2 + bx + c$
Graph	Straight line	Parabola (U-shaped)
Degree	1	2
Rate of Change	Constant	Variable
Vertex	None	Yes
Symmetry	None	Axis of symmetry
Max/Min	None (unless $m = 0$)	Max or Min at vertex
Domain	$\mathbb{R}$	$\mathbb{R}$
Range	$\mathbb{R}$	$[k, \infty)$ or $(-\infty, k]$

Comparison: Graphs Side by Side

Linear: $y = 2x + 1$



Quadratic: $y = x^2 - 2x - 3$



Shifting of Graphs: Introduction

Graph transformations allow us to sketch functions quickly by shifting, stretching, or reflecting a basic (parent) function.

Basic Transformations

Given a function $y = f(x)$ and a constant $c > 0$:

- **Vertical Shift:**
  - $y = f(x) + c$: Shift **up** by c units.
  - $y = f(x) - c$: Shift **down** by c units.
- **Horizontal Shift:**
  - $y = f(x - c)$: Shift **right** by c units.
  - $y = f(x + c)$: Shift **left** by c units.

Remember: Horizontal shifts are **opposite** to the sign inside the function.

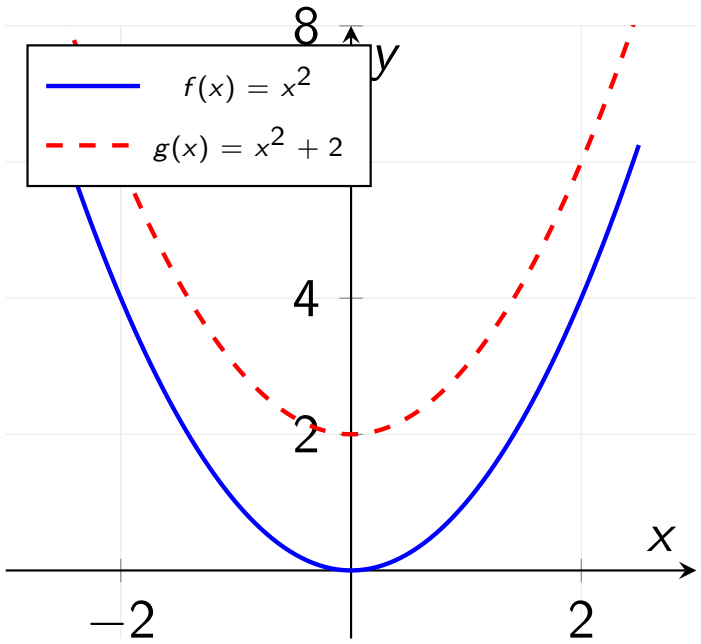
Example 1: Vertical Shift

Graph $f(x) = x^2$ **and** $g(x) = x^2 + 2$.

- $f(x) = x^2$ (parent function).
- $g(x) = x^2 + 2$: Shift **up** by 2 units.
- Every y-value increases by 2.

Key Points:

x	x^2	$x^2 + 2$
-2	4	6
-1	1	3
0	0	2
1	1	3
2	4	6



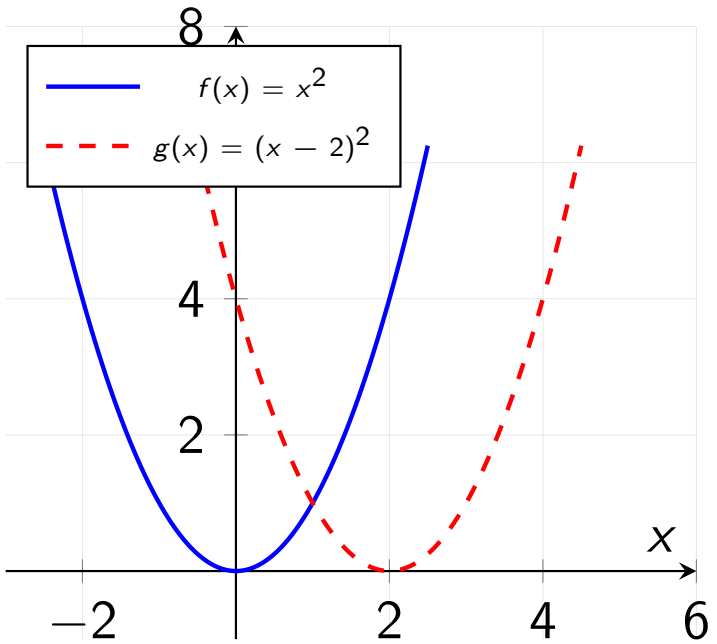
Example 2: Horizontal Shift

Graph $f(x) = x^2$ **and** $g(x) = (x-2)^2$.

- $f(x) = x^2$ (parent function).
- $g(x) = (x-2)^2$: Shift **right** by 2 units.
- Vertex moves from $(0, 0)$ to $(2, 0)$.

Key Points:

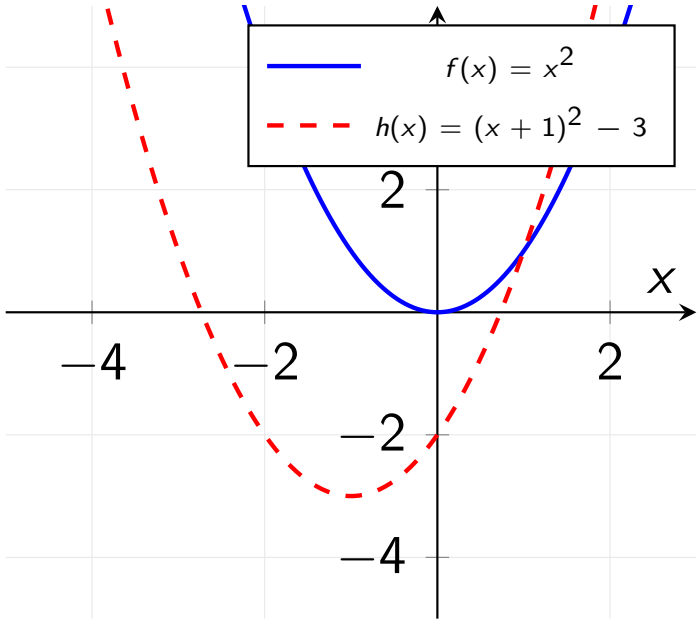
x	x^2	$(x-2)^2$
0	0	4
1	1	1
2	4	0
3	9	1
4	16	4



Example 3: Combined Shifts

Graph $f(x) = x^2$ **and** $h(x) = (x+1)^2 - 3$.

- Start with $f(x) = x^2$.
- $(x+1)^2$: Shift **left** by 1 unit.
- Then -3: Shift **down** by 3 units.
- Vertex: $(-1, -3)$.



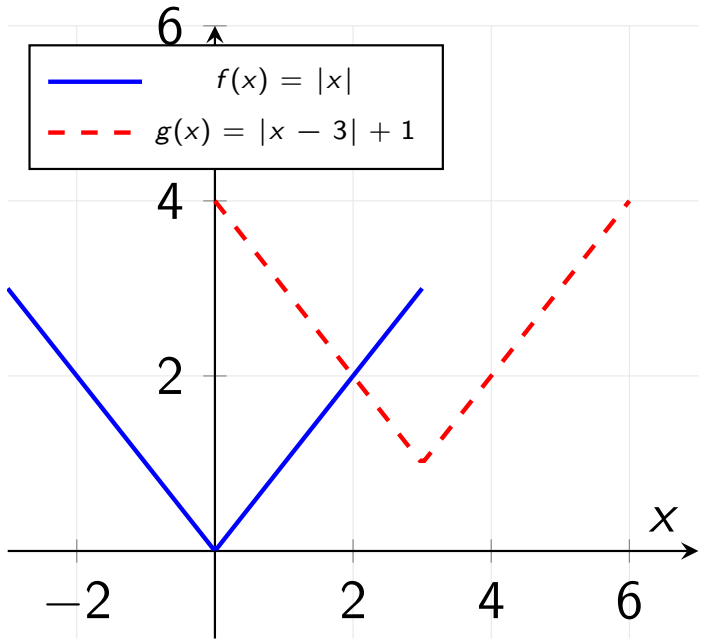
Key Points:

x	$(x+1)^2 - 3$
-3	1
-2	-2
-1	-3
0	-2
1	1

Example 4: Shifting the Absolute Value Function

Graph $f(x) = |x|$ **and** $g(x) = |x - 3| + 1$.

- Parent: $f(x) = |x|$, vertex at $(0, 0)$.
- $|x - 3|$: Shift **right** by 3 units.
- $+1$: Shift **up** by 1 unit.
- New vertex: $(3, 1)$.



Key Points:

x	$\lvert x - 3 \rvert + 1$
1	3
2	2
3	1
4	2
5	3

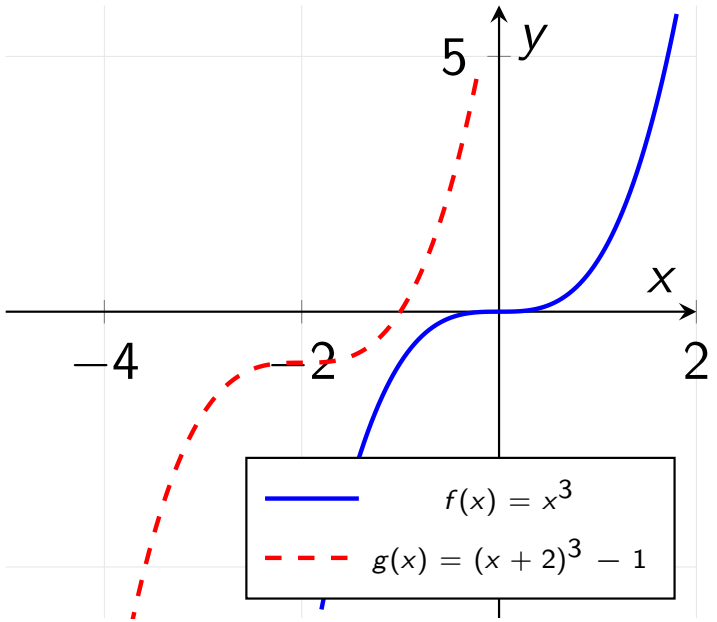
Example 5: Shifting a Cubic Function

Graph $f(x) = x^3$ **and** $g(x) = (x+2)^3 - 1$.

- Parent: $f(x) = x^3$.
- $(x+2)^3$: Shift **left** by 2 units.
- -1: Shift **down** by 1 unit.
- Inflection point: $(-2, -1)$.

Key Points:

x	$(x+2)^3 - 1$
-4	-9
-3	-2
-2	-1
-1	0
0	7



Summary of Graph Transformations

Transformation	**Equation**	**Effect**
Vertical Shift Up	$y = f(x) + c$	Up c units
Vertical Shift Down	$y = f(x) - c$	Down c units
Horizontal Shift Right	$y = f(x - c)$	Right c units
Horizontal Shift Left	$y = f(x + c)$	Left c units
Vertical Stretch	$y = c \cdot f(x)$, $c > 1$	Stretch vertically
Vertical Compression	$y = c \cdot f(x)$, $0 < c < 1$	Compress vertically
Reflection in x-axis	$y = -f(x)$	Flip over x-axis
Reflection in y-axis	$y = f(-x)$	Flip over y-axis

Tip

Always perform horizontal shifts and reflections **inside** the function first, then vertical shifts and stretches.

Conclusion

Key Takeaways

- A **function** maps each input to exactly one output.
- **Domain** = set of valid inputs; **Range** = set of possible outputs.
- The **graph** of a function is the set of all points $(x, f(x))$.
- **Linear functions** ($y = mx + b$) produce straight lines.
- **Quadratic functions** ($y = ax^2 + bx + c$) produce parabolas.
- **Graph transformations** (shifts, stretches, reflections) allow quick sketching from a parent function.

Thank You!